

Erfan Moeini

FULL-STACK MACHINE LEARNING ENGINEER

Aachen, Germany

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Personal Profile

Results-driven Machine Learning Engineer with an M.Sc. in Data Science from RWTH Aachen University and industry experience at Volkswagen, Mercedes-Benz, Amadeus, and INFORM. Specialized in end-to-end ML systems—time-series forecasting, AutoML, meta-learning, and MLOps. Lead technical development of Cognimata, an agentic AI automation platform. Skilled in building scalable systems with cloud-native and DevOps practices, with focus on generative AI, LLM integration, and automation-first engineering.

Work Experience

Artavision UG

Aachen, Germany

Lead Technical Engineer

Mar 2026 - Present

- Launched and developed Cognimata, a platform for AI-powered workflow automation and agentic AI systems targeting business process optimization.
- Provide AI consultancy services, advising organizations on AI strategy, LLM integration, and the adoption of automation-first engineering practices.

INFORM GmbH

Aachen, Germany

Software Developer (Optimization and AI)

Jun 2025 - Present

- Designed and deployed the department's MLOps platform, establishing end-to-end pipelines for model training, versioning, monitoring, and deployment.
- Developed AI-driven optimization software for container terminal logistics, improving scheduling and distribution efficiency across terminal operations.
- Contributed to the core Terminal Operating System (TOS), integrating machine learning modules to enhance real-time decision-making in port logistics.

Volkswagen AG

Wolfsburg, Germany

Machine Learning Research Intern

Jul 2024 - Jun 2025

- Developed and evaluated meta-learning strategies for automated deep learning model selection in automotive time-series forecasting as part of the R&D team.
- Benchmarked transfer learning and few-shot learning approaches across multiple time-series datasets, demonstrating improved model adaptation under limited labeled data.

PADS — Process and Data Science, RWTH Aachen

Aachen, Germany

Scientific Assistant

Mar 2024 - Jul 2024

- Contributed to the AIStudyBuddy project, a federal-state initiative combining Process Mining, Learning Analytics, and AI to support evidence-based study planning across three universities.
- Applied process mining to analyze student study behavior from campus and examination systems, comparing real study paths against intended curricula to derive actionable insights.

Amadeus Leisure IT GmbH

Kohlscheid, Germany

Software Developer

Apr 2023 - Jun 2024

- Designed and developed RESTful microservices using OpenAPI specification to aggregate and deliver travel data across multiple destination endpoints.
- Automated CI/CD pipelines with Jenkins and deployed containerized services on Kubernetes, reducing manual deployment overhead and improving release reliability.

Mercedes-Benz AG

Sindelfingen, Germany

Process Mining Intern

Sep 2023 - Dec 2023

- Engineered an end-to-end data pipeline for detecting and monitoring Overall Equipment Effectiveness (OEE) across manufacturing production lines.
- Developed interactive visualization dashboards for real-time downtime analysis, enabling production managers to identify and address performance bottlenecks.

Derak Cloud Ltd.

Shiraz, Iran

Data/Cloud Engineer

May 2021 - Oct 2022

- Developed and maintained a multi-tenant SaaS platform providing CDN, IaaS, and Video-on-Demand services to enterprise customers.
- Architected data pipelines using message queues and brokers for real-time data replication and infrastructure monitoring.
- Built a real-time tweet processing and sentiment analysis application enabling data-driven advertising decisions by the marketing team.

Deed Asia Ltd.
AI Intern

- Developed a License Plate Recognition (LPR) system combining OCR with YOLO-based object detection, trained on a custom-labeled dataset of Iranian license plates.

Shiraz, Iran
Jul 2019 - Sep 2019

Education

RWTH Aachen University
Data Science M.Sc.

- Thesis: Meta-Learning Approaches for Time-Series Deep Learning (conducted at Volkswagen AG).
- Ranked among the top 5% of students based on academic performance.

Aachen, Germany
Oct 2022 - Mar 2025

Shiraz University
Computer Engineering B.Sc.

- Ranked 851st in the National Entrance Exam (among 150,000 applicants).
- Finished with an 18.54/20 CGPA. Ranked 3rd among the 81 students of my class.

Shiraz, Iran
Sep 2018 - Aug 2022

Academic Projects

Performance Spectrum and Correlation Plot Widgets
RWTH Aachen University - Process Mining in Celonis Lab

- Designed and implemented interactive widgets on the Celonis Platform for performance spectrum visualization and correlation analysis, simplifying bottleneck detection and root cause analysis in business processes.

Aachen, Germany
Winter 2022/2023

Aphrodite, A Covid-19 Detection Application
Shiraz University - Systems Analysis and Design Course

- Built and deployed a dockerized Covid-19 detection web application using the pre-trained COVIDNet-CT model, developed in collaboration with Shiraz University of Medical Sciences.

Shiraz, Iran
Fall 2021

Real-time Pose Estimation
Shiraz University - Open Research Project

- Estimated cable crossover machine force using 3D pose estimation by integrating TRT-Pose on NVIDIA Jetson Nano for real-time inference at the edge.

Shiraz, Iran
Fall 2021

Skills

Machine Learning & AI	PyTorch, TensorFlow, Keras, Scikit-Learn, Ray, Pandas, NumPy, SciPy, MLflow, Metaflow
Agentic Frameworks/Workflows	LangChain, LangGraph, Crew AI, OpenAI Agents SDK, Claude Agents SDK, n8n
Programming Languages	Python, C/C++, Java, JavaScript, Go, R
Data Engineering	Spark, Kafka, Hadoop, Prometheus, Grafana, ELK Stack
Cloud & DevOps	Docker, Kubernetes, Jenkins, Argo CD, Ansible, CI/CD, Azure ML, CUDA
Databases	MySQL, MongoDB, ClickHouse, Firebase, Redis
Web Frameworks & APIs	OpenAPI, Django, Flask, Spring Boot, Node.js, Angular, Vue.js, D3.js
Tools & Platforms	Linux, Git, Shell (Bash/Zsh), $\LaTeX$, JIRA

Achievements

2023	Ranked First, Public Transport & Micro-mobility Competition , Hackathon	<i>Paris, France</i>
2023	Dean's List , RWTH Aachen University	<i>Aachen, Germany</i>
2020	Ranked First in the Technical Footsteps Competition , I.D.E.A Association	<i>Shiraz, Iran</i>

Languages

English	Full Proficiency
German	Intermediate Proficiency
Persian	Native Proficiency